Lecture 7. Trade with Increasing Returns to Scale and Monopolistic Competition

Lectures on International Trade

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Recall from Previous Lectures

- Trade between different countries
  - Ricardian model: different productivity
  - Specific-factors model: different factors or productivity
  - Heckscher-Ohlin model: different endowments
- Trade between similar countries
  - If two countries are the same, is there incentive to trade?
- In reality, most of the trade happens among those developed economies. Most of them are quite similar.
- Then, what explains these trade between these similar countries?
  - Monopolistic competition: a country import and export the same good, but different varieties.
  - In perfectly competitive markets, the goods produced are homogeneous. In this chapter, we assume that goods are differentiated, and allow for imperfect competition.

Outline

- Basics of Monopolistic Competition
- Trade under Monopolistic Competition
- Gains and Costs from Trade with Monopolistic Competition
- NAFTA
- Gravity Equation

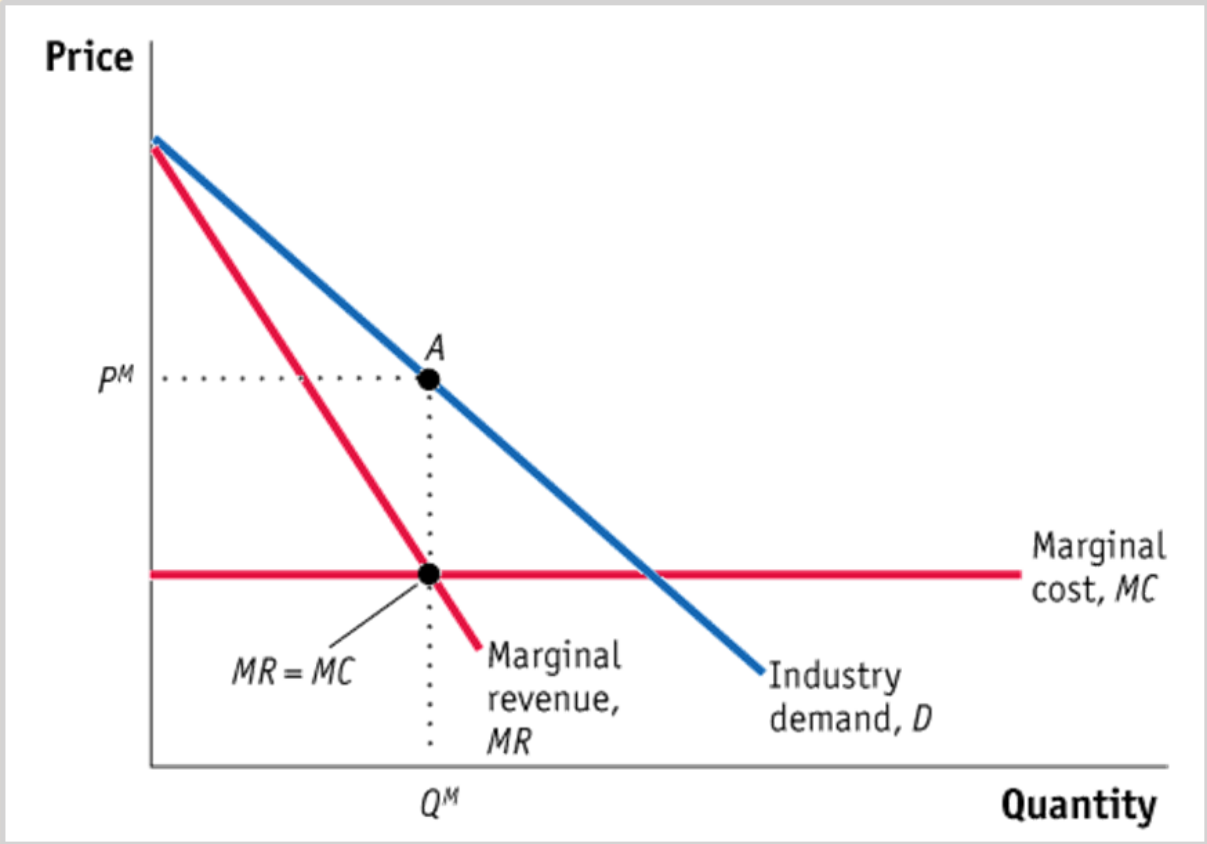
Monopolistic Competition

- Monopolistic competition has two key features:
  - The goods produced by different firms are differentiated.
  - Firms enjoy increasing returns to scale, by which we mean that the average costs for a firm fall as more output is produced.
- Marginal revenue: the extra or marginal revenue the firm gains from selling an additional unit.
- Marginal cost: cost of producing an additional unit.

Monopolistic Competition, cont'

Monopoly Equilibrium: Profit Maximization

FIGURE 6-1



Monopoly Equilibrium

The monopolist chooses the profit-maximizing quantity, Q^M, at which marginal revenue equals marginal cost.

From that quantity, we trace up to the demand curve and over to the price axis to see that the monopolist charges the price P^M.

The monopoly equilibrium is at point A.

Monopolistic Competition, cont'

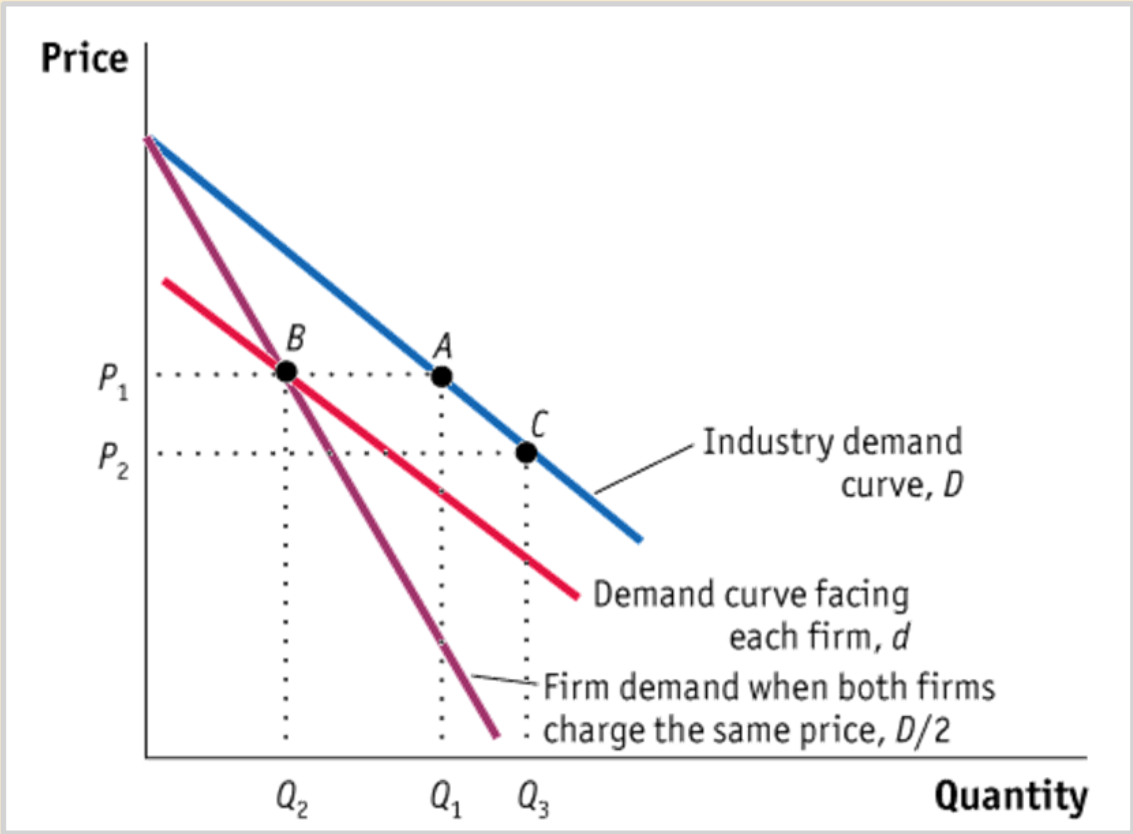
Monopoly Equilibrium: Example

- Demand: $Q = 200 - P$
- Total Cost: $C = 50 + 40Q$;
- So MC: $MC = 40$
- Total revenue: $\Pi = PQ = (200 - Q)Q$
- MR: $MR = \frac{\partial \Pi}{\partial Q} = 200 - 2Q$
- Equilibrium: MR=MC
  - $Q^M = 80$; $P^M = 120$

Monopolistic Competition, cont'

Demand with Duopoly

FIGURE 6-2 (1 of 2) **Demand Curves with Duopoly**

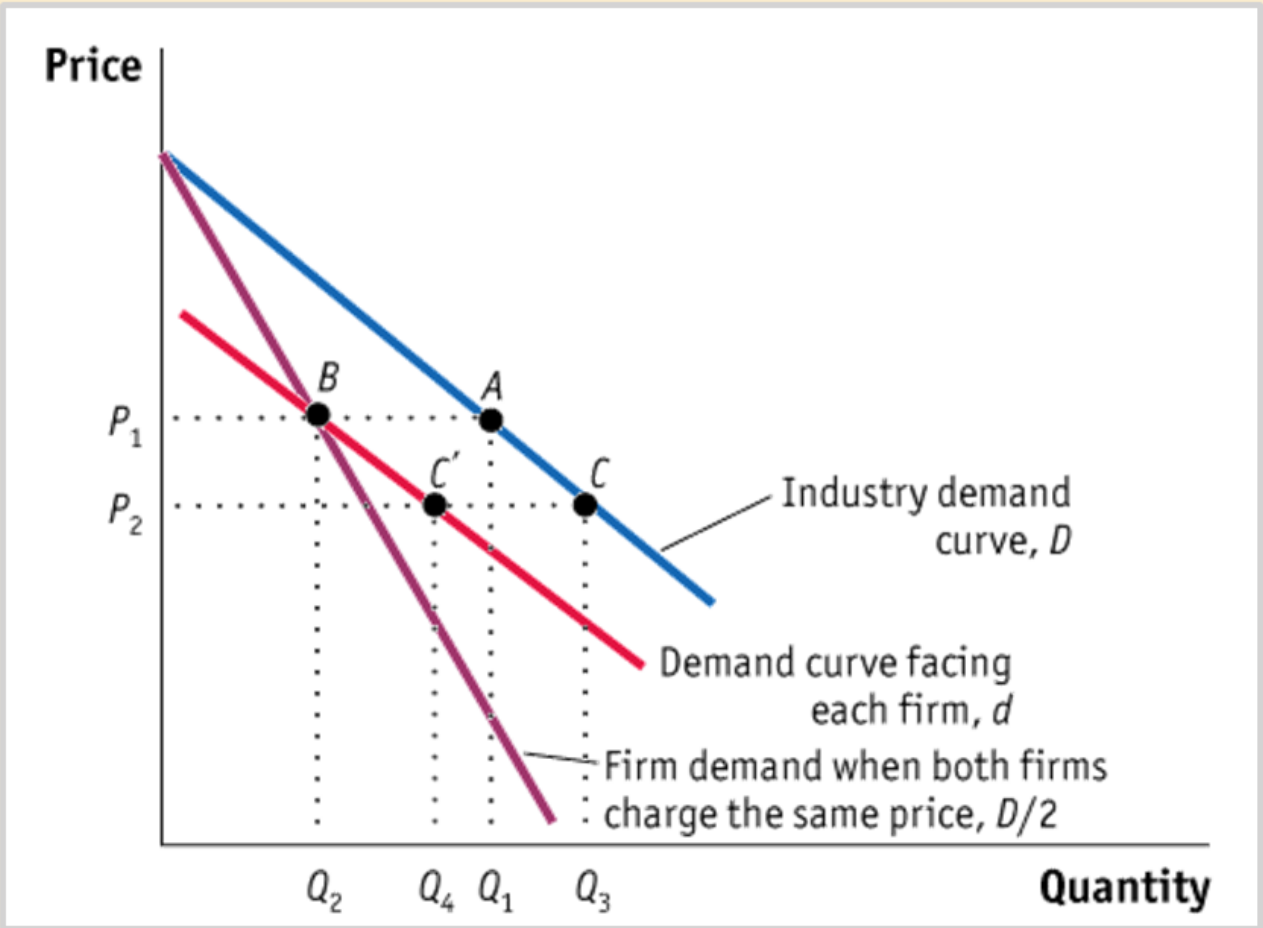


When there are two firms in the market and they both charge the same price, each firm faces the demand curve $D/2$. At the price P_1, the industry produces Q_1 at point A and each firm produces $Q_2 = Q_1/2$ at point B. If both firms produce identical products and one firm lowers its price to P_2, all consumers will buy from that firm only; the firm that lowers its price will face the demand curve, D, and sell Q_3 at point C.

Monopolistic Competition, cont'

Demand with Duopoly

FIGURE 6-2 (2 of 2) **Demand Curves with Duopoly**



Alternatively, if the products are differentiated, the firm that lowers its price will take some, but not all, sales from the other firm; it will face the demand curve, d, and at P_2 it will sell Q_4 at point **C'**.

Trade under Monopolistic Competition

Assumptions of the model of monopolistic competition:

- **Assumption 1**: Each firm produces a good that is similar to but slightly differentiated from the goods that other firms in the industry produce.
- Each firm faces a downward-sloping demand curve for its product and has some control over the price it charges.
- **Assumption 2**: There are many firms in the industry.
- If the number of firms is N, then D/N is the share of demand that each firm faces when the firms are all charging the same price.
- When only one firm lowers its price, however, it will face a flatter demand curve d.

Trade under Monopolistic Competition, cont'

- **Assumption 3**: Firms produce using a technology with increasing returns to scale.
- Increasing returns to scale cause average costs to fall as the quantity produced increases.
- There might be many reasons for increasing returns to scale
  - Fixed cost of production
  - Specialization

Trade under Monopolistic Competition, cont'

Numerical Example of Increasing Returns to Scale

TABLE 6-2

Cost Information for the Firm **This table illustrates increasing returns to scale, in which average costs fall as quantity rises.**

Quantity Q	Variable Costs = $Q \cdot MC$ (MC = \$10)	Total Costs = Variable Costs + Fixed Costs (FC = \$100)	Average Costs = Total Costs/Quantity
10	\$100	\$200	\$20
20	200	300	15
30	300	400	13.3
40	400	500	12.5
50	500	600	12
100	1,000	1,100	11
Large Q	10 • Q	10 • Q + 100	Close to 10

Whenever the price charged is above average costs, then a firm earns monopoly profits.

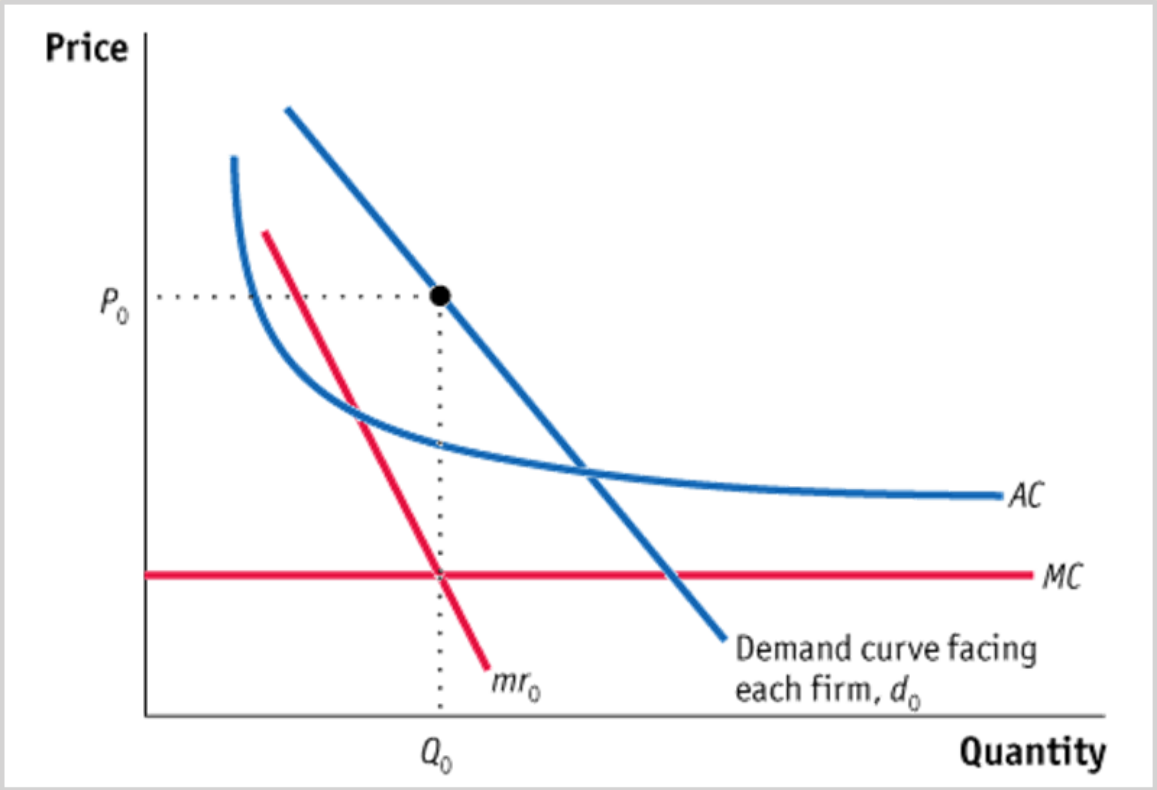
Trade under Monopolistic Competition, cont'

- **Assumption 4**: Because firms can enter and exit the industry freely, monopoly profits are zero in the long run.
- Firms will enter as long as it is possible to make monopoly profits, and the more firms that enter, the lower profits per firm become.
- Profits for each firm end up as zero in the long run, just as in perfect competition.
- Next we are going to examine monopolistic competition
  - in both **short-run** and **long-run** (Note that the difference between short- and long-run is there will be zero profit in the long run. In the short run, firm may earn positive profit.)
  - **with** and **without** trade

Trade under Monopolistic Competition, cont'

Short-Run Equilibrium without Trade

FIGURE 6-4 Short-Run Monopolistic Competition Equilibrium without Trade



The short-run equilibrium under monopolistic competition is the same as a monopoly equilibrium.

The firm chooses to produce the quantity Q_0 at which the firm's marginal revenue, mr_0, equals its marginal cost, MC.

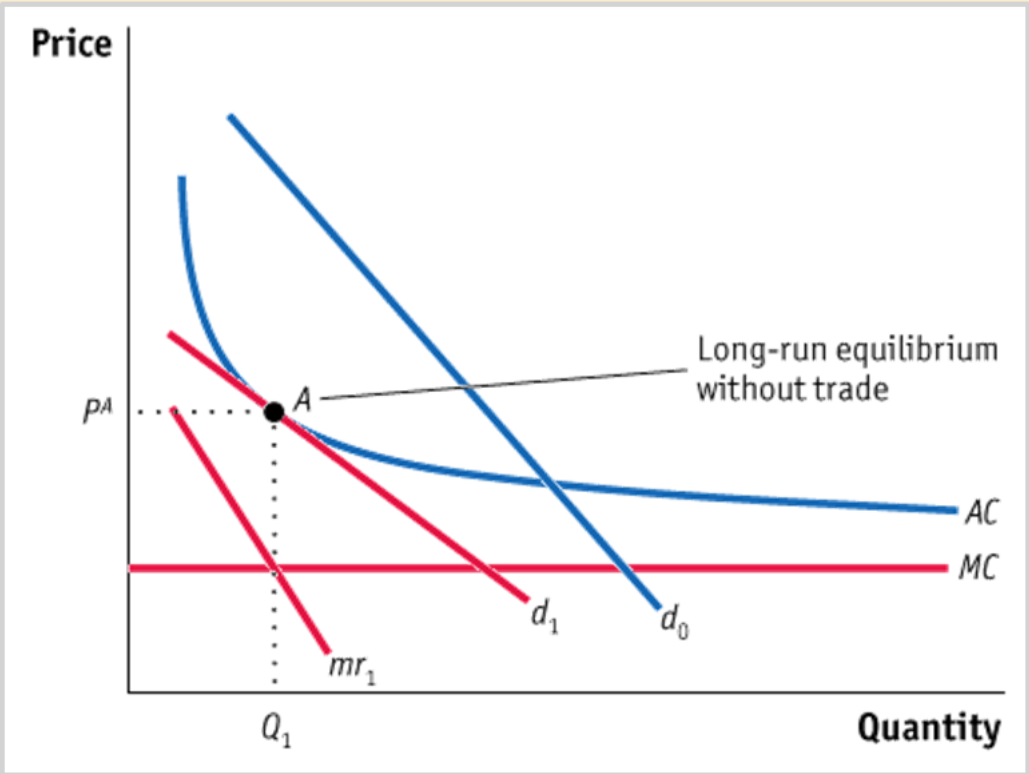
The price charged is P_0.

Because price exceeds average cost, the firm makes monopolistic profit.

Trade under Monopolistic Competition, cont'

Long-Run Equilibrium without Trade

FIGURE 6-5 (1 of 2) Long-Run Monopolistic Competition Equilibrium without Trade



Drawn by the possibility of making profits in the short-run equilibrium, new firms enter the industry and the firm's demand curve, d_0, shifts to the left and becomes more elastic (i.e., flatter), shown by d_1.

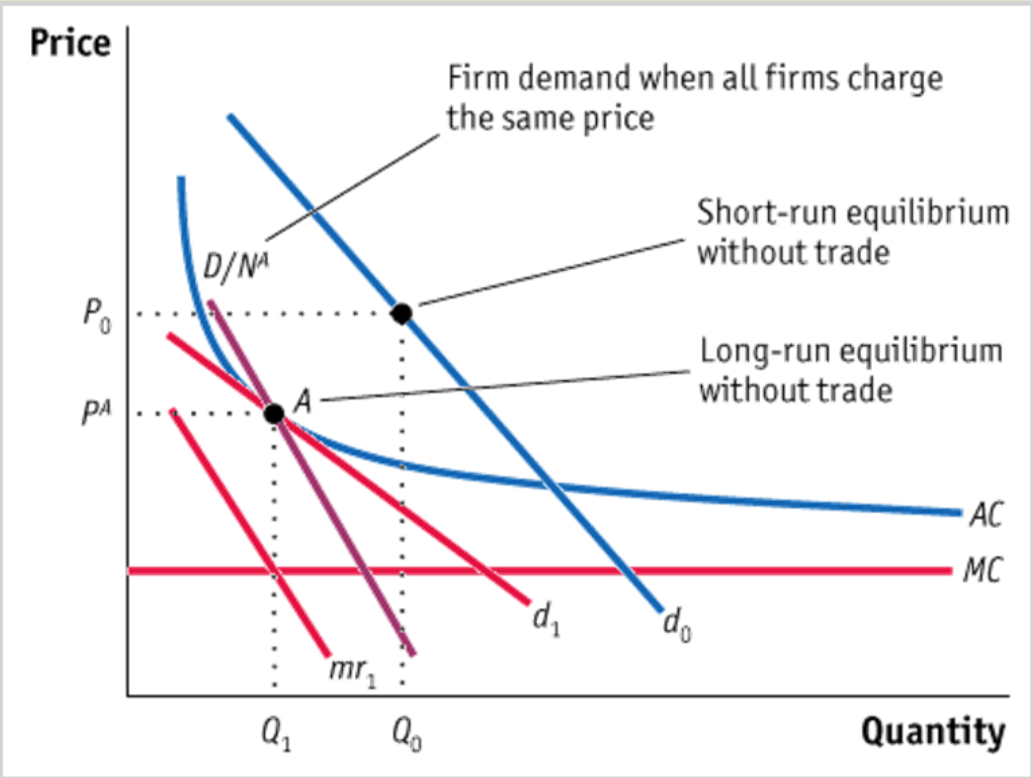
The long-run equilibrium under monopolistic competition occurs at the quantity Q_1 where the marginal revenue curve, mr_1 (associated with demand curve d_1), equals marginal cost.

At that quantity, the no-trade price, P^A, equals average costs at point A.

Trade under Monopolistic Competition, cont'

Long-Run Equilibrium without Trade

FIGURE 6-5 (2 of 2) **Long-Run Monopolistic Competition Equilibrium without Trade**



In the long-run equilibrium, firms earn zero monopoly profits and there is no entry or exit. The quantity produced by each firm is less than in short-run equilibrium (Figure 6-4). Q_1 is less than Q_0 because new firms have entered the industry.

With a greater number of firms and hence more varieties available to consumers, the demand for each variety d_1 is less then d_0. The demand curve D/N^A shows the no-trade demand when all firms charge the same price.

Trade under Monopolistic Competition, cont'

Equilibrium with Free Trade

- Assume Home and Foreign are exactly the same.
  - Same number of consumers
  - Same technology and cost curves
  - Same number of firms in the no-trade equilibrium
- Given the above conditions, if there were no economies of scale, there would be no reason for trade.
  - Under the Ricardian model, countries with identical technologies would not trade.
  - Under the Heckscher-Ohlin model, countries with identical factor endowments would not trade.
- However, under monopolistic competition, two identical countries will still engage in trade.

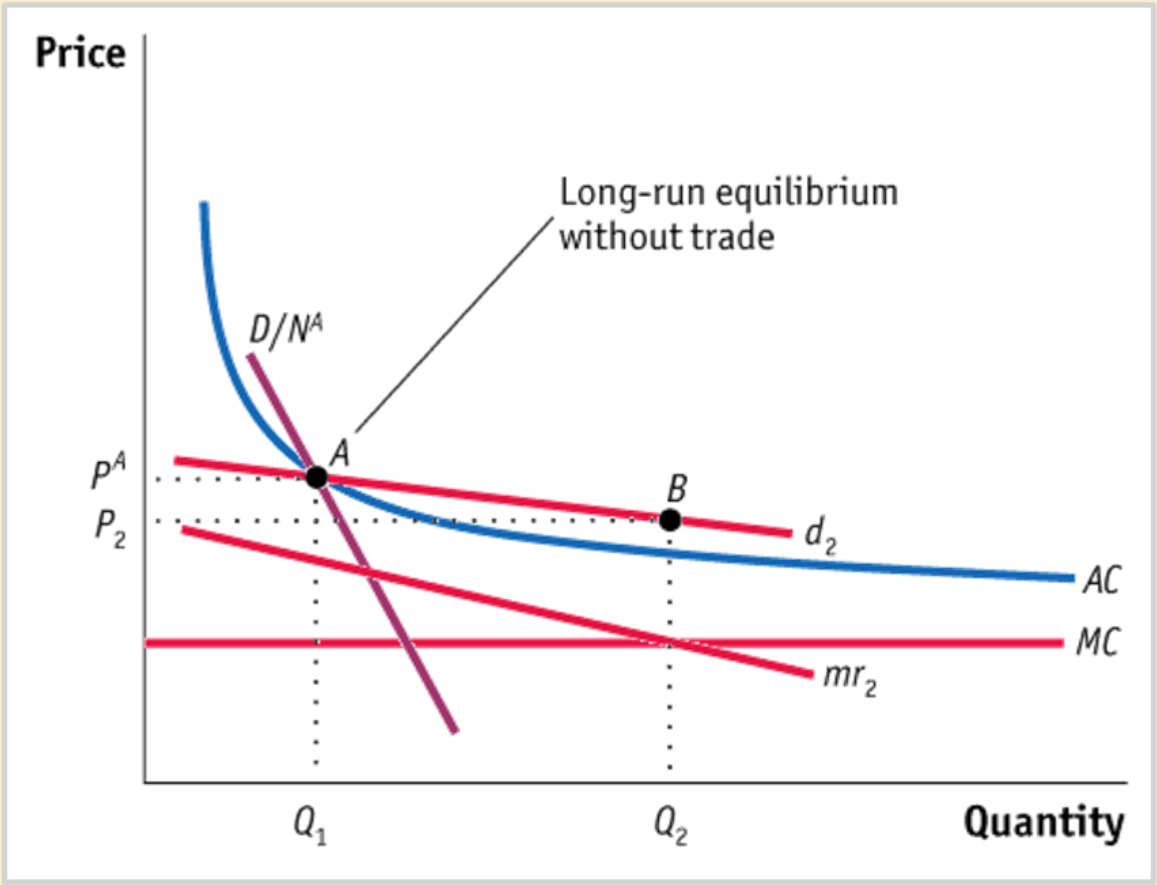
Trade under Monopolistic Competition, cont'

- The number of firms in the no-trade equilibrium in each country is N^A.
- First, we will consider each country in long-run equilibrium without trade
- When trade opens, the number of customers available to each firm doubles.
- Since there are twice as many consumers, but also twice as many firms, the ratio stays the same.
- The product varieties also double.
- With the greater number of varieties available, the demand for each individual variety will be more elastic.

Trade under Monopolistic Competition, cont'

Short-Run Equilibrium with Trade

FIGURE 6-6 (1 of 2) **Short-Run Monopolistic Competition Equilibrium with Trade**



When trade is opened, the larger market makes the firm's demand curve more elastic, as shown by d_2 (with corresponding marginal revenue curve, mr_2).

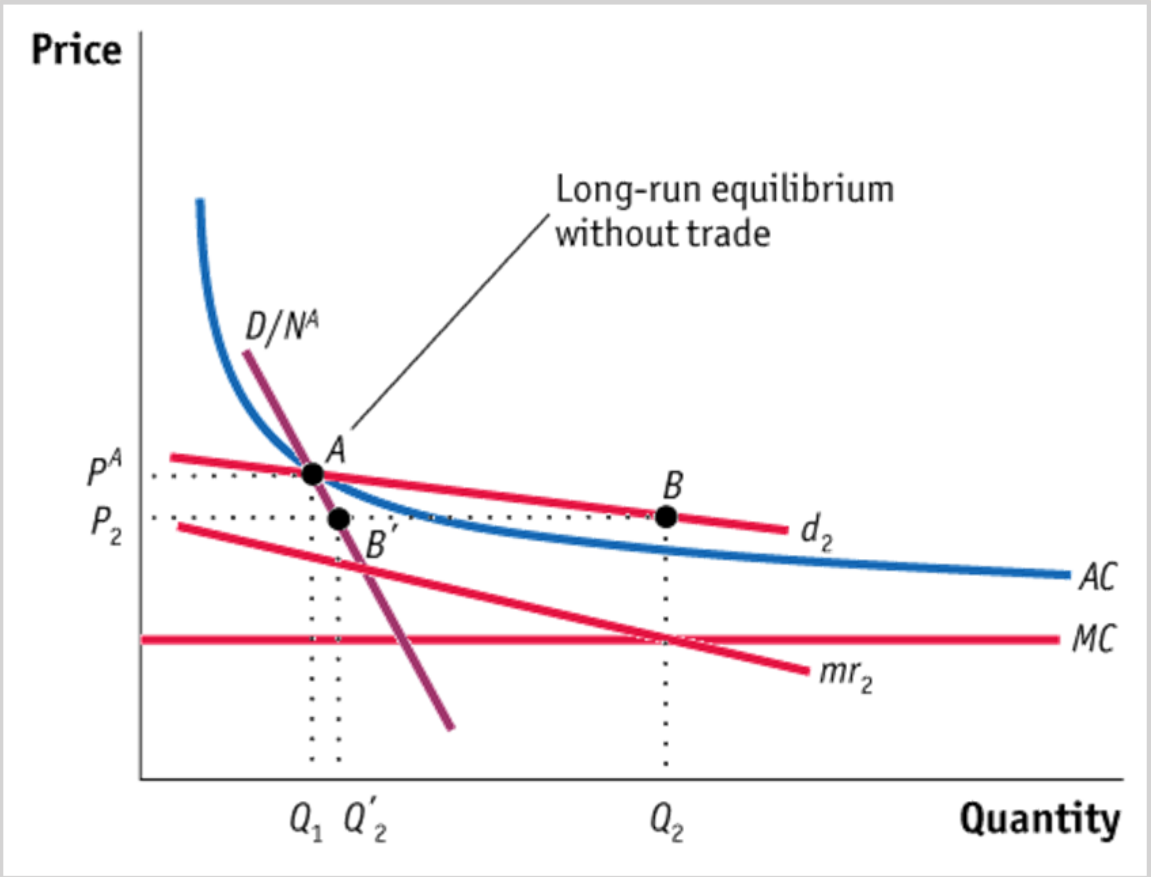
The firm chooses to produce the quantity Q_2 at which marginal revenue equals marginal costs;

this quantity corresponds to a price of P_2. With sales of Q_2 at price P_2, the firm will make monopoly profits because price is greater than *AC*.

Trade under Monopolistic Competition, cont'

Short-Run Equilibrium with Trade

FIGURE 6-6 (2 of 2) **Short-Run Monopolistic Competition Equilibrium with Trade**



When *all* firms lower their prices to P_2, however, the relevant demand curve is D/N^A, which indicates that they can sell only Q'_2 at price P_2.

At this short-run equilibrium (point *B'*), price is less than average cost and all firms incur losses. As a result, some firms are forced to exit the industry.

Trade under Monopolistic Competition, cont'

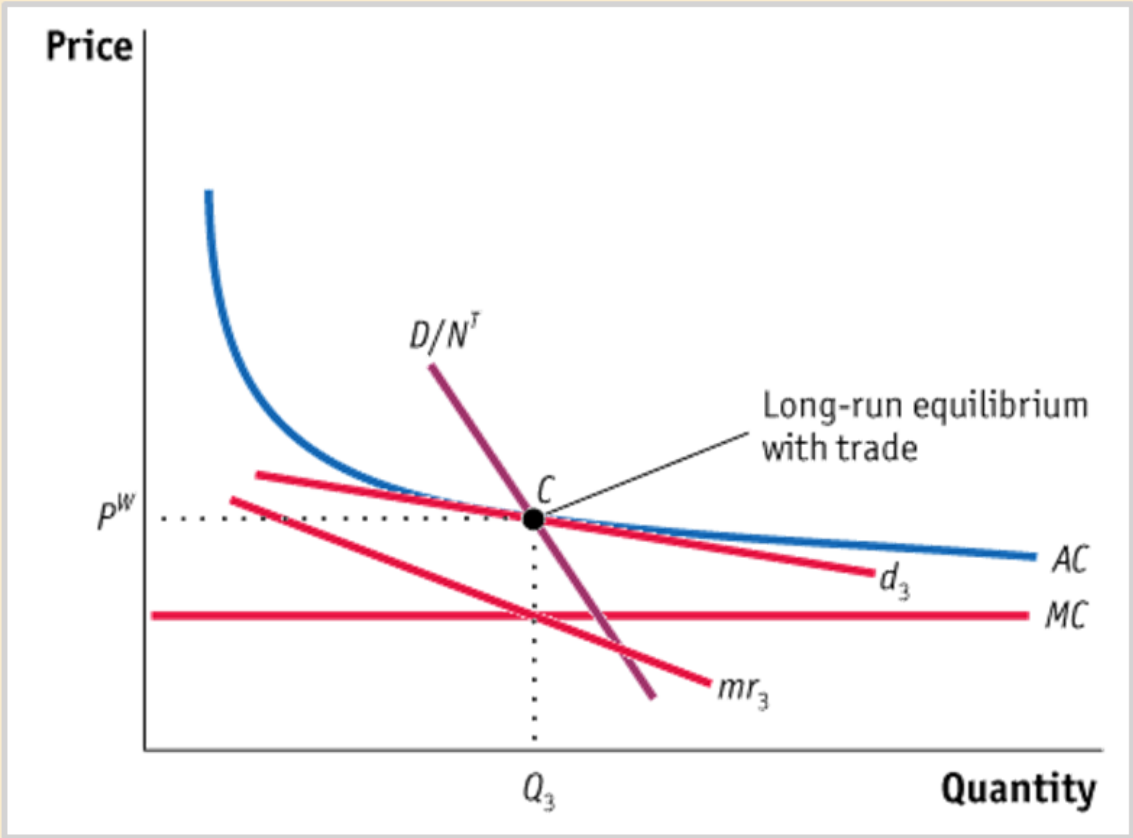
Long-Run Equilibrium with Trade

- Since firms are making losses, some of them will exit the industry.
- Firm exit will increase demand for the remaining firms' products and decrease the available product varieties (produced in each country) to consumers.
- We now have N^T firms which is fewer than the N^A firms we had before.
- The new demand D/N^T lies to the right of D/N^A.

Trade under Monopolistic Competition, cont'

Long-Run Equilibrium with Trade

FIGURE 6-7 (1 of 2) **Long-Run Monopolistic Competition Equilibrium with Trade**



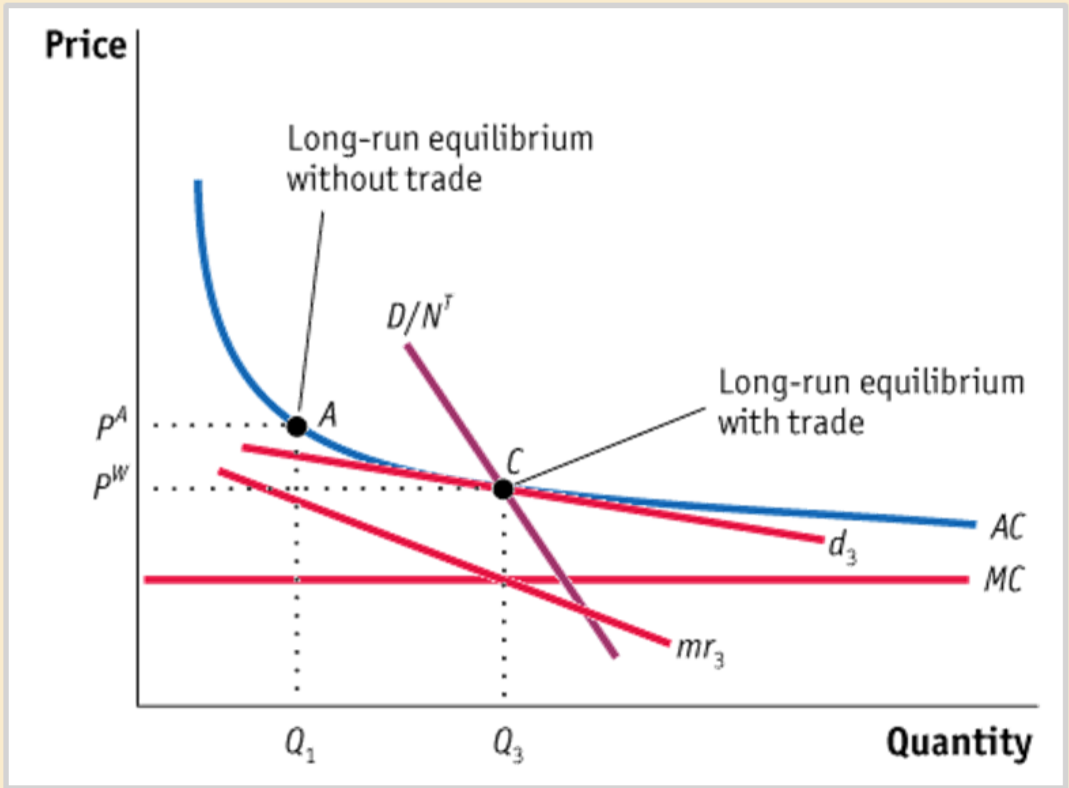
The long-run equilibrium with trade occurs at point *C*.

At this point, profits are maximized for each firm producing Q_3 (which satisfies $mr_3 = MC$) and charging price P^W (which equals *AC*). Since monopoly profits are zero when price equals average cost, no firms enter or exit the industry.

Trade under Monopolistic Competition, cont'

Long-Run Equilibrium with Trade

FIGURE 6-7 (2 of 2) **Long-Run Monopolistic Competition Equilibrium with Trade (continued)**



Compared with the long-run equilibrium without trade (Figure 6-5), d_3 (along with mr_3) has shifted out as domestic firms exited the industry and has become more elastic due to the greater total number of varieties with trade, $2N^T > N^A$.

Compared with the long-run equilibrium without trade at point *A*, the trade equilibrium at point *C* has a lower price and higher sales by all surviving firms.

Gains from Trade under Monopolistic Competition

The long-run equilibrium at point C has **two sources of gains** from trade for consumers:

- A drop in price.
  - The lower price is a result of increased productivity of the surviving firms coming from increasing returns to scale.
- Gains from trade to consumers.
  - Although there are fewer product varieties made within each country (by fewer firms), consumers have more product variety because they can choose products of the firms from both countries after trade.

Trade under Monopolistic Competition, cont'

Adjustment Costs from Trade

- There are adjustment costs associated with monopolistic competition, as some firms shut down or exit the industry.
- Workers in those firms experience a spell of unemployment.
- Over the long run, however, we could expect those workers to find new jobs, so these costs are temporary.
- We will examine both short-run and long-run adjustment costs.
- Next, we look at evidence from Mexico, Canada, and the United States following the North American Free Trade Agreement (NAFTA).

Trade under Monopolistic Competition, cont'

Gains and Costs from Trade in Canada after NAFTA

- In Canada, there were very large initial declines in employment.
- Over time, however, these job losses were more than made up for by the creation of new jobs elsewhere in manufacturing.
- Productivity growth in Canada allowed for a modest rise in real earnings.

Studies of NAFTA on the U.S. economy have not estimated its effects on the productivity of U.S. firms. Among the reasons is that Mexico and Canada are only two of many trading partners with the U.S.

Trade under Monopolistic Competition, cont'

Gains and Adjustment Costs for Mexico under NAFTA

- NAFTA resulted in a decrease in tariffs. How did the fall in tariffs affect the Mexican economy?
- NAFTA also increased the productivity of the maquiladora plants over and above the increase in productivity that occurred in the rest of Mexico.
- For workers, however, there was a fall of more than 20% in real wages in both manufacturing and agriculture, despite a rise in productivity.

Trade under Monopolistic Competition, cont'

Gains and Adjustment Costs for Mexico under NAFTA

- Farmers growing corn in Mexico did not suffer as much as was feared.
  - The poorest farmers can always consume the corn they grow, rather than sell it.
  - The Mexican government also applied subsidies to offset the reduction in income for corn farmers.
- The total production of corn in Mexico actually rose after NAFTA.
- The maquiladoras face increasing international competition (not all due to NAFTA), which can be expected to raise the volatility of its output and employment.

Trade under Monopolistic Competition, cont'

Summary of NAFTA

- For the U.S., the long-run gains have consisted of an expansion of varieties, and a fall in consumer prices.
- It is clear that for Canada and the U.S., the long-run gains considerably exceed the short-run costs.
- In Mexico, the gains have not translated into the growth of real wages for workers.
- However, the real earnings of higher-income workers in the maquiladora sector have risen. They have been the principal beneficiaries of NAFTA so far.

Intra-Industry Trade and the Gravity Equation

Index of Intra-Industry Trade:

The **index of intra-industry trade** tells us what proportion of trade in each product involves both imports and exports: a high index (up to 100%) indicates that an equal amount of the good is imported and exported, whereas a low index (0%) indicates that the good is either imported or exported but not both.

$$\text{(index of intra-industry trade for good i)} = \frac{\min\{\text{Import}_i, \text{Export}_i\}}{\frac{1}{2}(\text{Import}_i + \text{Export}_i)}$$

The Gravity Equation

The Gravity Equation

- Dutch economist and Nobel laureate, Jan Tinbergen, was trained in physics and thought of comparing the trade between countries to the force of gravity between objects.
- In physics, objects with a larger mass, or those that are close together, have greater gravitational pull between them.
- In economics, the gravity equation for trade states that countries with larger GDPs, or that are close to each other, will have more trade between them.

The Gravity Equation, cont

The Gravity Equation

- The force of gravity between these two masses is:

$$F = G\frac{m_1 \cdot m_2}{r^2}$$

- where m is the mass, r is the distance
- The Gravity Equation in Trade:

$$T = B\frac{GDP_1 \cdot GDP_2}{d^n}$$

- where T is total trade between two countries, d is the distance between two countries.

$$T = \frac{1}{GDP_W}\frac{GDP_1 \cdot GDP_2}{d^n}$$

The Gravity Equation, cont'

The Gravity Term

- If trade across borders happens to be less than trade within countries, there must be barriers to trade between those countries.
- Factors that make it easier or more difficult to trade goods between countries are often called border effects, and they include the following:
  - Tariffs: Taxes imposed when imported goods enter into a country
  - Quotas: Limits on the number of items allowed to cross the border
  - Other administrative rules and regulations affecting trade, including the time required for goods to clear customs
  - Geographic factors such as whether the countries share a border
  - Cultural factors such as whether the countries have a common language that might make trade easier

Summary

Key Points:

1. When firms have differentiated products and increasing returns to scale, there is a potential for gains from trade that did not exist in earlier models.

2. The model of monopolistic competition shows that trade will occur between countries even if these countries are identical.

3. The monopolistic competition model assumes differentiated products, many firms, and increasing returns to scale. Firms enter whenever there are profits to be earned, so profits are zero in the long-run equilibrium.

Summary (cont)

4. When trade opens between two countries, the demand curve faced by each firm becomes more elastic, as consumers have more choices and become more price-sensitive. Firms then lower their prices in an attempt to capture consumers from their competitors and obtain profits. When all firms do so, however, some firms incur losses and are forced to leave the market.

5. Introducing international trade under monopolistic competition leads to additional gains from trade for two reasons: (i) lower prices as firms expand their output and lower their average costs and (ii) additional imported product varieties available to consumers. There are also short-run adjustment costs, such as unemployment, as some firms exit the market.

Summary (cont)

6. The assumption of differentiated goods helps us to understand why countries often import and export varieties of the same type of good. That outcome occurs within the model of monopolistic competition.

7. The gravity equation states that countries with higher GDP, or that are close, will trade more. In addition, research has shown that there is more trade within countries than between countries.